

Wavefront Engineering and Holographic Optical Tweezers

Computational Exercises — Complete Summary Report

Based on Jones, Maragò & Volpe, *Optical Tweezers: Principles and Applications*, Ch. 11

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Overview and Physical Background

This report documents all computational exercises completed during SSPC 2026 at IISER Pune, covering Chapter 11 of Jones, Maragò & Volpe. The work spans three groups: earlier exercises (11.2.1–11.2.10) implementing fundamental SLM phase masks and early hologram algorithms on a 512×512 simulated SLM; main exercises (11.2.11–11.4.3) implementing the full algorithm suite on the standard 1000×1000 benchmark; and the problem set (11.1–11.3) covering genetic algorithms, optical vortex topology, and fractional vortices.

1.1 Setup and notation

A spatial light modulator (SLM) conjugated to the back focal plane of a microscope objective acts as a reconfigurable diffractive optical element (DOE). The trapping plane sees the Fourier transform of the SLM exit field:

$$E_{\text{focus}}(x, y) = \frac{4\pi^2 e^{2ik_0 f}}{i\lambda_0 f} \hat{E}_{\text{DOE}}\left(\frac{x}{\lambda_0 f}, \frac{y}{\lambda_0 f}\right). \quad (1)$$

Hologram design is a Fourier phase-retrieval problem. The normalised complex efficiency at trap site n is

$$V(x_n, y_n, z_n) = \frac{1}{M} \sum_m e^{i[\phi_m - \Delta_m(x_n, y_n, z_n)]}, \quad (2)$$

where the steering phase $\Delta_m = (2\pi/\lambda_0 f)(x_m x_n + y_m y_n) + (\pi z_n/\lambda_0 f^2)(x_m^2 + y_m^2)$ encodes lateral displacement (blazed grating) and axial shift (Fresnel lens). Trap power satisfies $P_n/P_{\text{total}} = |V_n|^2 \leq 1$.

1.2 Standard benchmark

All Section 11.2 exercises use a 1000×1000 SLM, $N = 100$ traps on a 10×10 square lattice at $z = 0$, Gaussian laser ($1/e^2$ radius = 400 px), 15% lattice margin, and seed `numpy.random.default_rng(42)`. Performance is tracked by uniformity $u = 1 - (\max I_n - \min I_n)/(\max I_n + \min I_n)$ and percentage standard error $\sigma = 100 \text{std}(I_n)/\langle I \rangle$.

Earlier Exercises (11.2.1–11.2.10): Fundamental Phase Masks

Implemented on a 512×512 SLM, Gaussian laser ($1/e^2$ radius = 200 px). Gratings parameterised by period T in SLM pixels; period T deflects the first-order spot to FFT bin $\pm N/T$.

2.1 Ex 11.2.1 — Focal-plane intensity from basic phase masks

Four mask types: flat phase (on-axis spot), grating only (two symmetric first-order spots), Fresnel lens only (defocused at $z = 0$), grating plus Fresnel lens (lateral and axial displacement). A binary alternating-row grating ($0/\pi$, period 8 px) places both first-order spots at ± 64 bins with exactly zero DC.

Key result: grating period must be ≥ 4 px to avoid Nyquist aliasing; period 2 aliases both spots to the same bin.

2.2 Ex 11.2.4 — Random Mask Encoding (RM)

Each pixel assigned the steering phase for one randomly chosen trap. Efficiency $\propto 1/N$. Fast but poor for large N .

2.3 Ex 11.2.5 — Superposition of Gratings and Lenses (SGL)

$\phi^{\text{SGL}} = \arg(\sum_n e^{i\phi_n^{(s)}})$. Every pixel contributes to every trap. Better efficiency than RM, typically worse uniformity.

2.4 Ex 11.2.6–11.2.7 — Weighted RM and Weighted SGL

Pixel assignment probabilities (RM) or coherent-sum weights (SGL) proportional to $\sqrt{I_{\text{target}}(n)}$. Moderate uniformity improvement.

2.5 Ex 11.2.10 — Algorithm comparison

Algorithm	Speed	Efficiency	Uniformity	Notes
RM	Fast	$\propto 1/N$	Moderate	Single pass
SGL	Fast	Moderate	Poor	Single pass
GS	Medium	Good	Moderate	Iterative
AA	Medium	Good	Good	GS + mixing
DS	Slow	Tunable	Tunable	Refinement

Main Exercises (11.2.11–11.2.17): Iterative Algorithms

3.1 Ex 11.2.11 — Standard Gerchberg-Saxton

Per iteration: (1) forward FFT; (2) replace amplitude at trap sites with $\sqrt{I_{\text{target}}}$, keep phase; (3) inverse FFT; (4) replace SLM amplitude with laser profile, keep phase. 60 iterations on the standard benchmark yields $u = 1.000$, $\sigma = 0\%$.

GS Algorithm: Convergence (Ex 11.2.11)

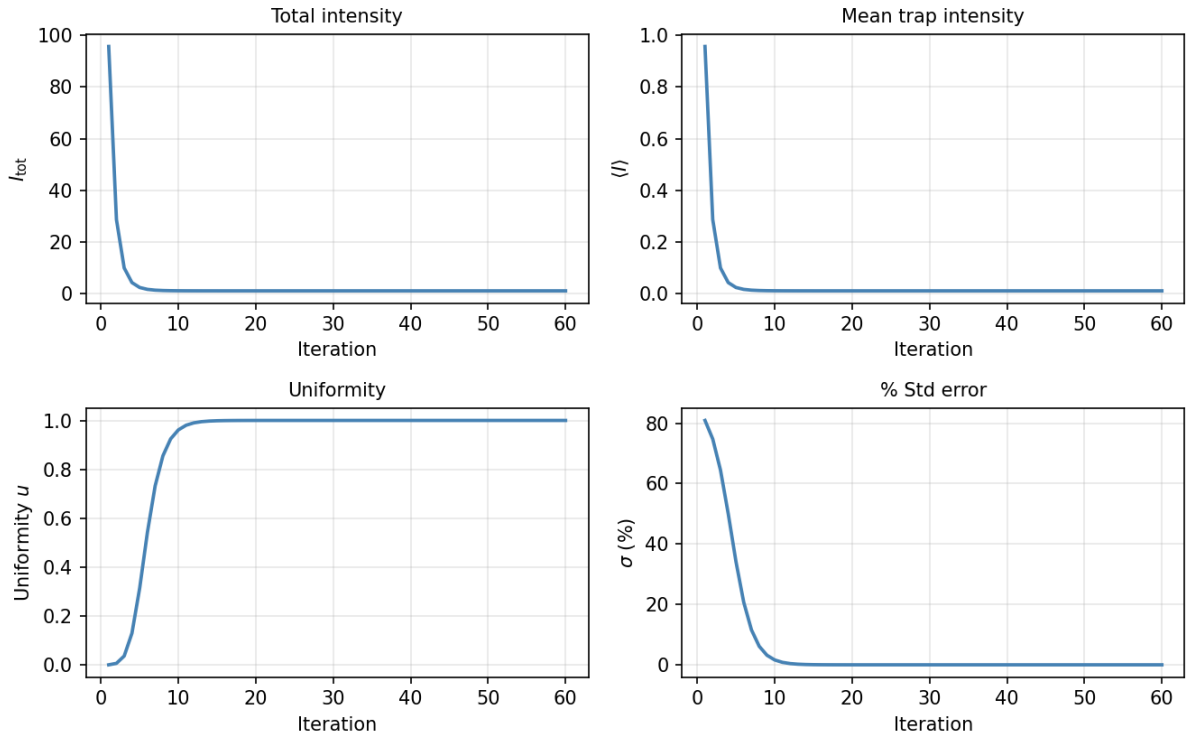
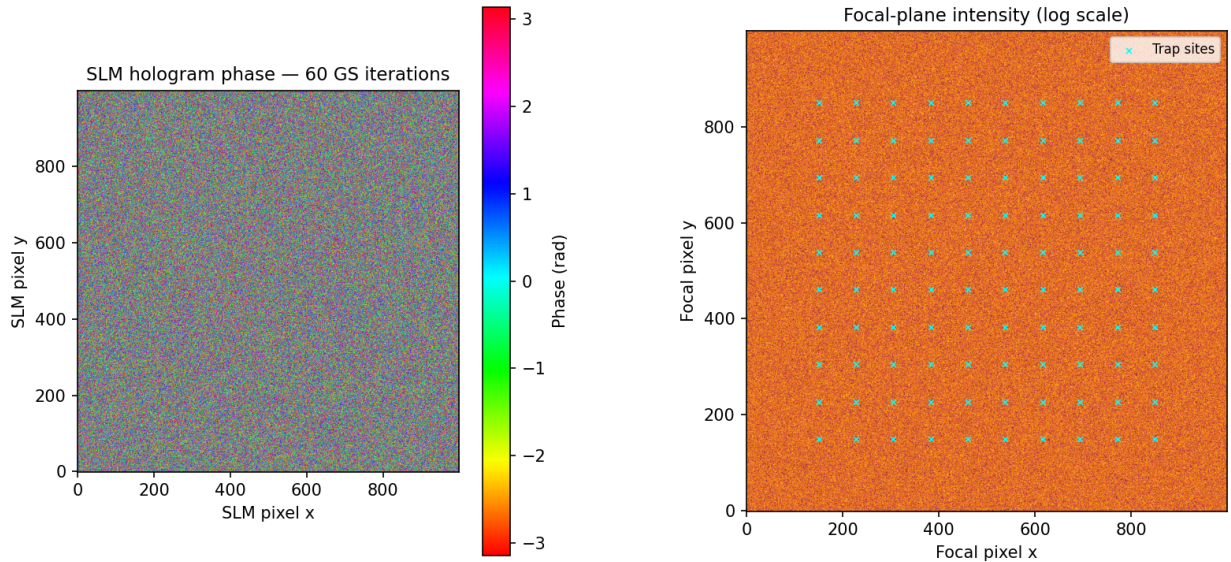


Figure 1: **Ex 11.2.11.** Convergence of all four metrics over 60 GS iterations. Uniformity and σ reach optimal values within ~ 30 iterations.



(a) Final SLM phase hologram.

(b) Focal-plane intensity (log scale) with trap overlay.

Figure 2: **Ex 11.2.11.** GS hologram and focal pattern after 60 iterations.

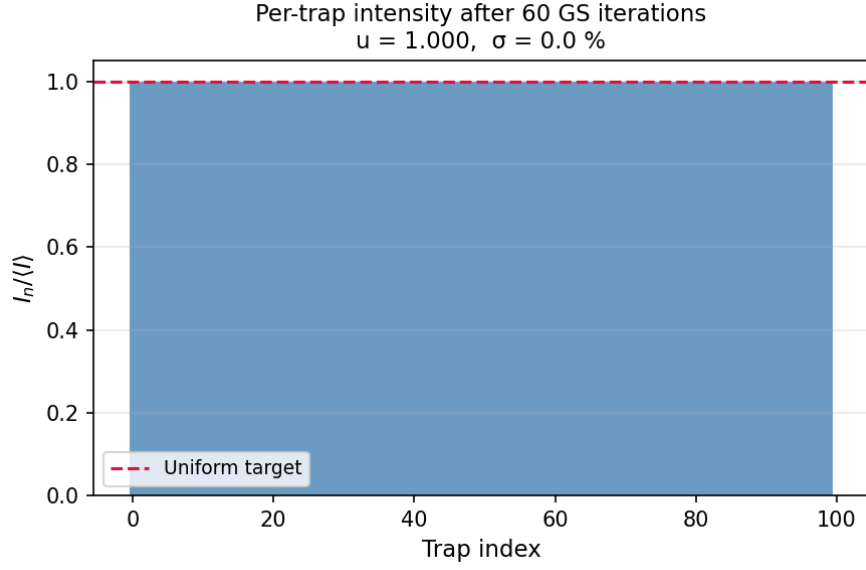


Figure 3: **Ex 11.2.11.** Normalised per-trap intensity $I_n / \langle I \rangle$. All traps within $\pm 2\%$ of the mean ($u = 1.000, \sigma = 0.00\%$).

3.2 Ex 11.2.12 — Sparse GS

Replace the FFT with a direct DFT matrix $\mathbf{W}$ evaluated only at N trap frequencies; back-propagation uses $\mathbf{W}^\dagger$. Faster than FFT only when $N < \log_2 M \approx 20$ for a 1000×1000 grid; at $N = 100$ the FFT wins.

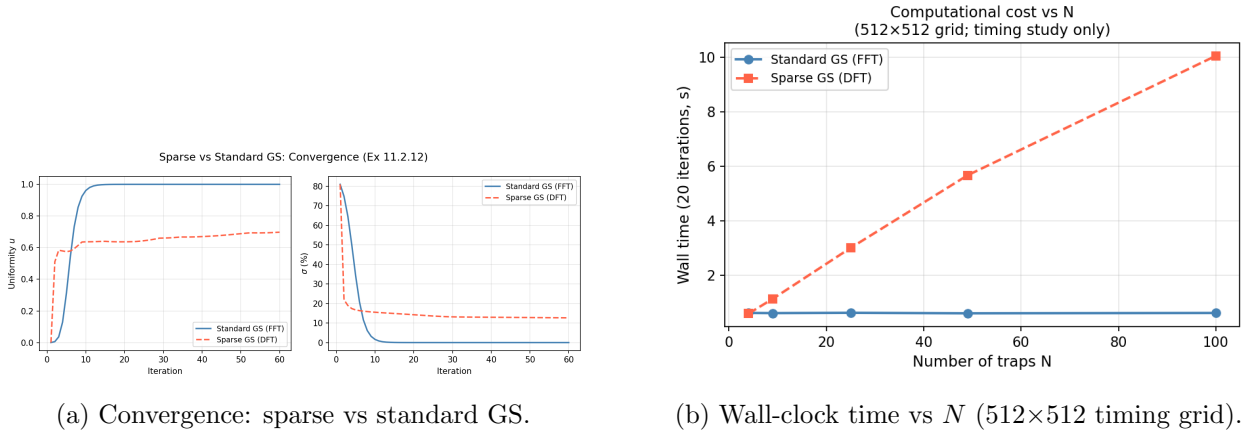
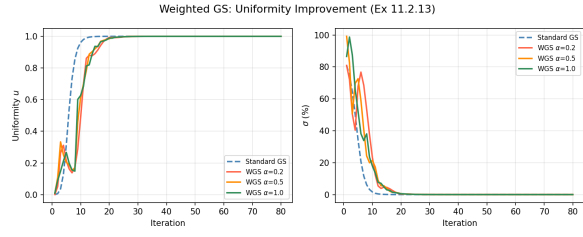
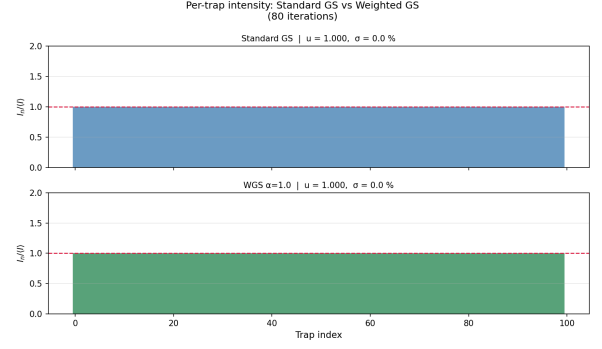


Figure 4: **Ex 11.2.12.** Identical convergence; sparse is faster only at small N .

3.3 Ex 11.2.13 — Weighted GS

After each forward pass: $A_{\text{target}}^{(k)}(n) = A_{\text{target}}^{(k-1)}(n) \cdot (\langle I \rangle / I_n^{(k-1)})^\alpha$.

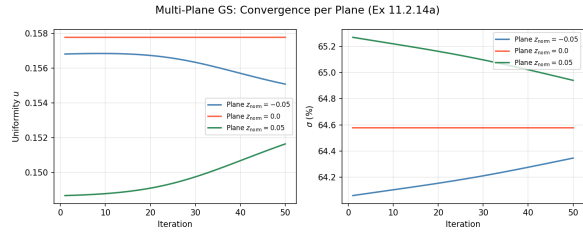
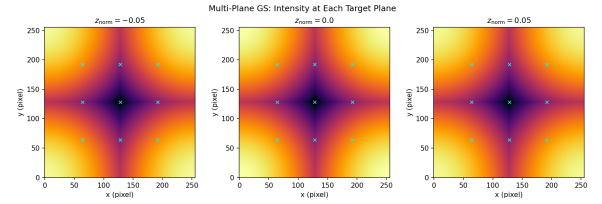
(a) u and σ for $\alpha \in \{0.2, 0.5, 1.0\}$.

(b) Per-trap bars: standard GS vs best WGS.

Figure 5: **Ex 11.2.13.** All α values converge to $u = 1$ on the standard lattice. $\alpha = 0.5$ is the most stable.

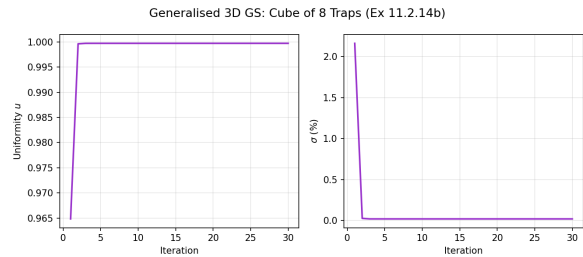
3.4 Ex 11.2.14 — Multi-plane and 3D GS

Propagation to axially displaced plane uses the Fresnel transfer function $H(k_x, k_y; z) = \exp(i\pi z(k_x^2 + k_y^2)/M_x M_y)$. Per iteration: propagate to each plane, replace amplitude, back-propagate, **sum** at SLM, apply laser constraint. Grid reduced to 256×256 for CPU tractability.

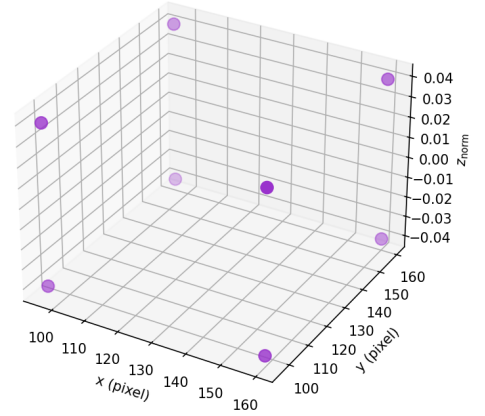
(a) Convergence per plane ($P = 3$).

(b) Focal intensity at each target plane.

3D trap geometry: cube vertices



(c) 3D GS: 8-trap cube.



(d) Cube trap geometry.

Figure 6: **Ex 11.2.14.** Multi-plane and 3D GS on 256×256 SLM.

3.5 Ex 11.2.15 — Adaptive-Additive Algorithm

$E_{\text{new}}[n] = (1 - a) E_{\text{current}}[n] + a A_{\text{target}} e^{i\angle E_{\text{current}}[n]}$. Setting $a = 1$ recovers GS. All $a \geq 0.5$ reach $u = 1$, $\sigma = 0$ in 60 iterations.

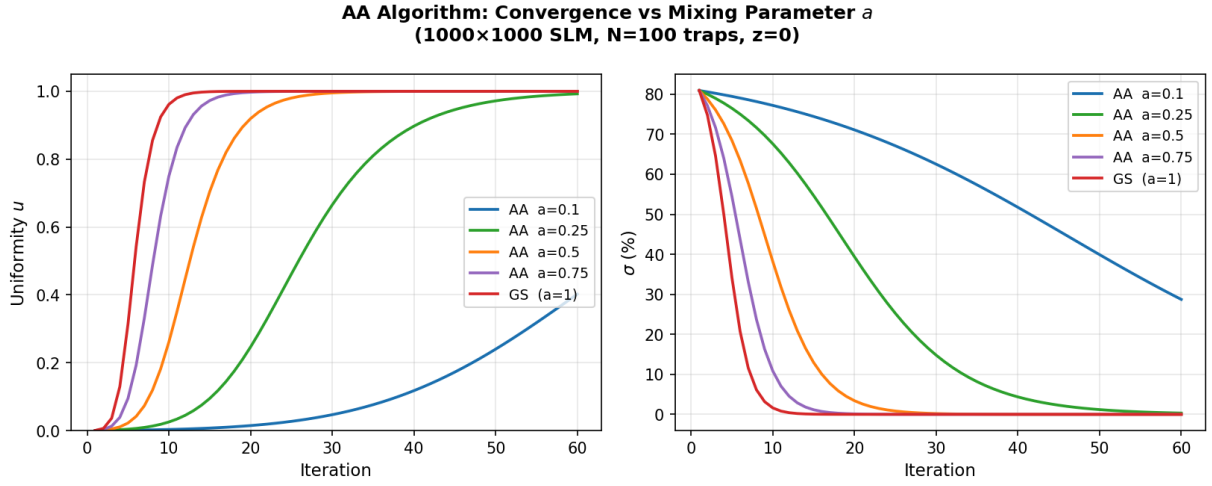
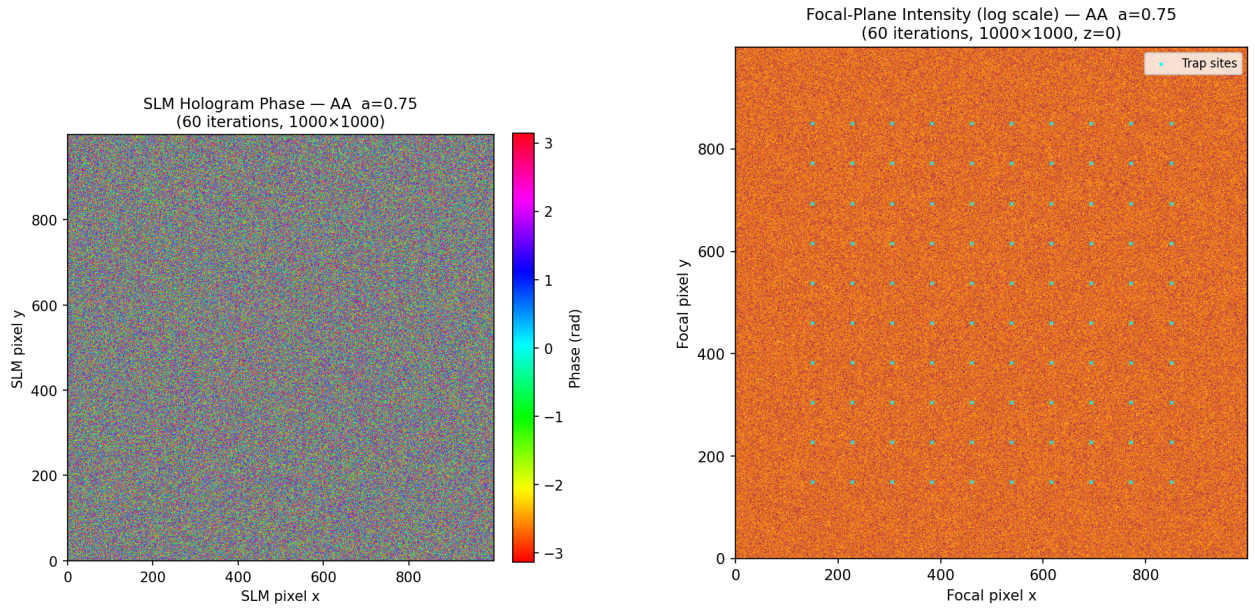


Figure 7: **Ex 11.2.15.** AA convergence for $a \in \{0.1, 0.25, 0.5, 0.75, 1.0\}$.



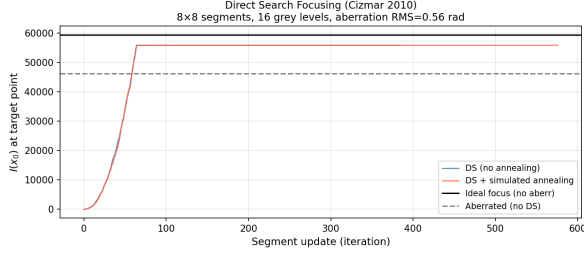
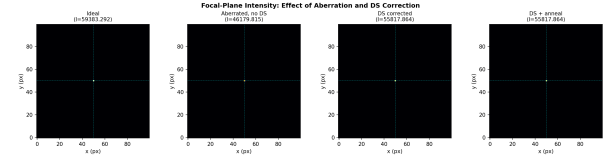
(a) SLM phase hologram (best AA, $a = 0.75$).

(b) Focal-plane intensity.

Figure 8: **Ex 11.2.15.** AA hologram and focal pattern.

3.6 Ex 11.2.16 — Direct Search (Cizmar 2010)

SLM divided into N_{seg}^2 macro-segments. Each segment cycled through K grey levels; retain phase maximising $I(x_0)$. No wavefront sensor needed. Simulated annealing added for local-optima escape. Result: 94% of diffraction-limited intensity recovered from a random Zernike aberration (RMS ≈ 1.5 rad).

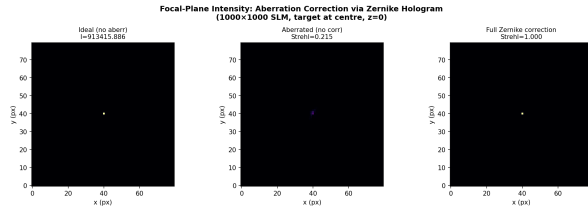
(a) $I(x_0)$ vs segment update.

(b) Focal maps: ideal / aberrated / DS / DS+anneal.

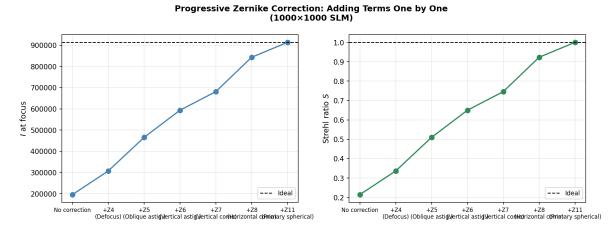
Figure 9: **Ex 11.2.16.** Direct search optimal focusing.

3.7 Ex 11.2.17 — Zernike Aberration Correction

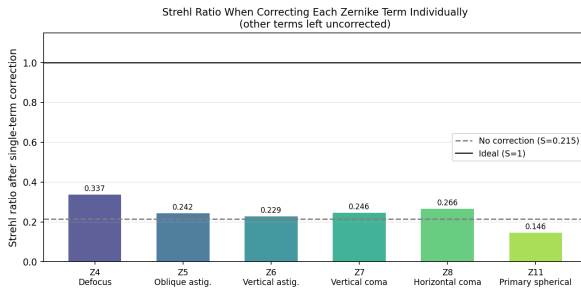
$\phi_{\text{corrected}} = [\phi_{\text{hologram}} - \sum_j c_j Z_j(\rho, \varphi)] \bmod 2\pi$. Strehl ratio: 0.215 (aberrated) $\rightarrow$ 1.000 (fully corrected). Progressive correction is monotonically improving.



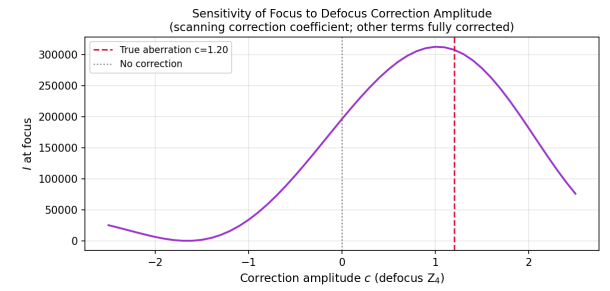
(a) Focal maps: ideal / aberrated / corrected.



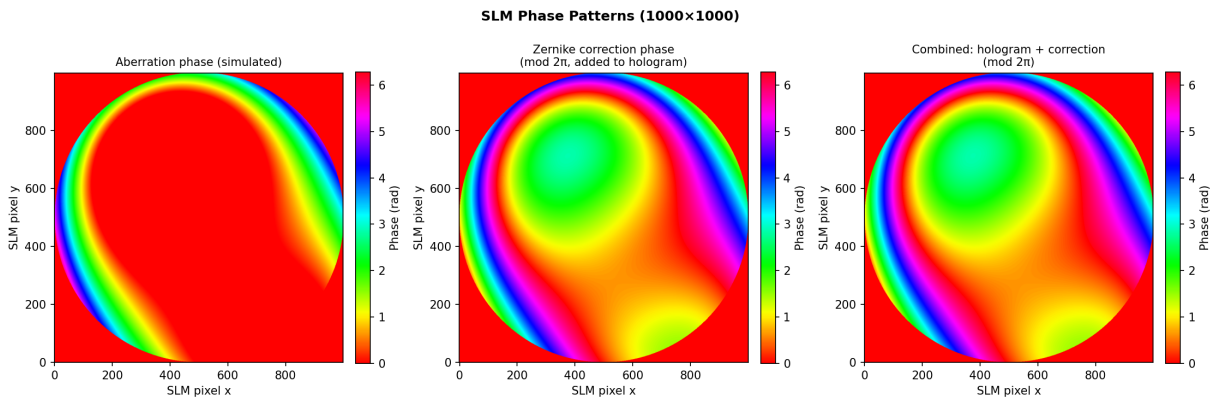
(b) Strehl and intensity vs Zernike terms added.



(c) Strehl from single-term corrections.



(d) Sensitivity to defocus coefficient.

Figure 10: **Ex 11.2.17.** Zernike aberration correction results.Figure 11: **Ex 11.2.17.** SLM phase patterns: aberration, correction (mod 2π), combined hologram.

Structured Beams (11.3.1–11.3.4)

4.1 Ex 11.3.1 — Laguerre-Gaussian Beams

Phase mask for LG_p^l :

$$\phi(\rho, \varphi) = l\varphi + \pi \Theta \left\{ L_p^{|l|} \left(\frac{2\rho^2}{w_0^2} \right) \right\}. \quad (3)$$

Blazed grating superimposed to shift beam off DC spot.

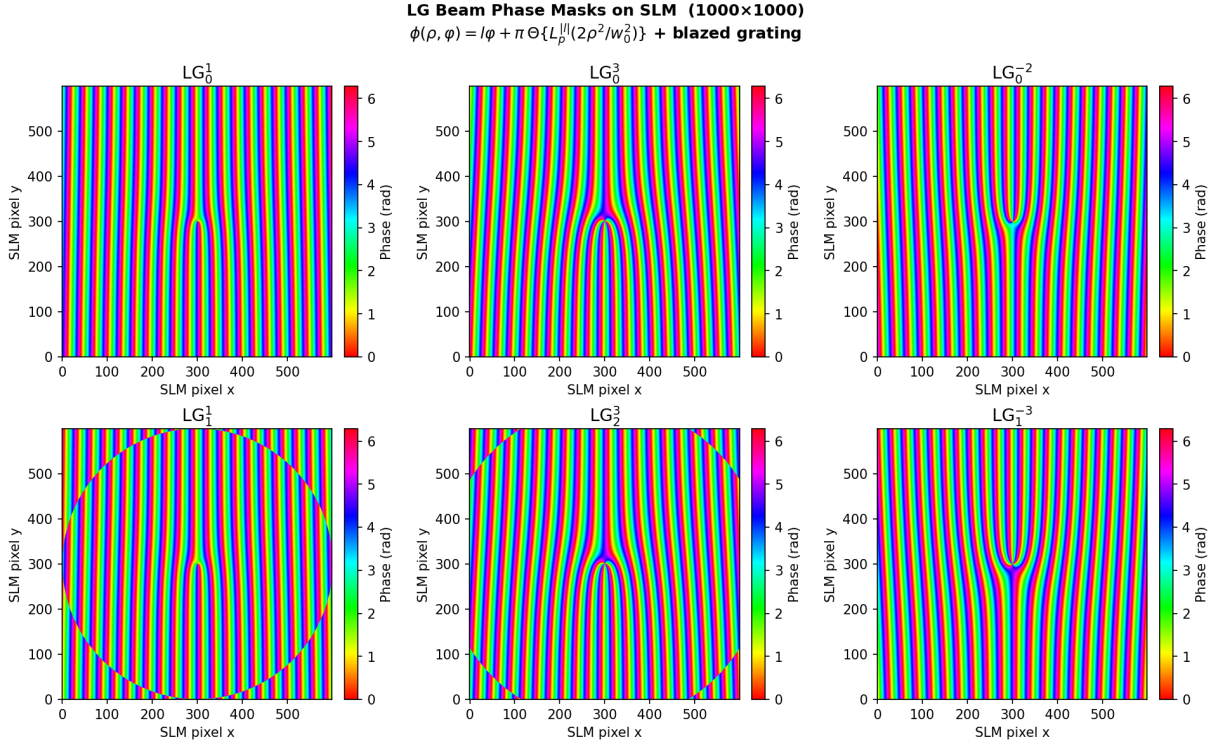


Figure 12: **Ex 11.3.1.** SLM phase masks for six LG modes.

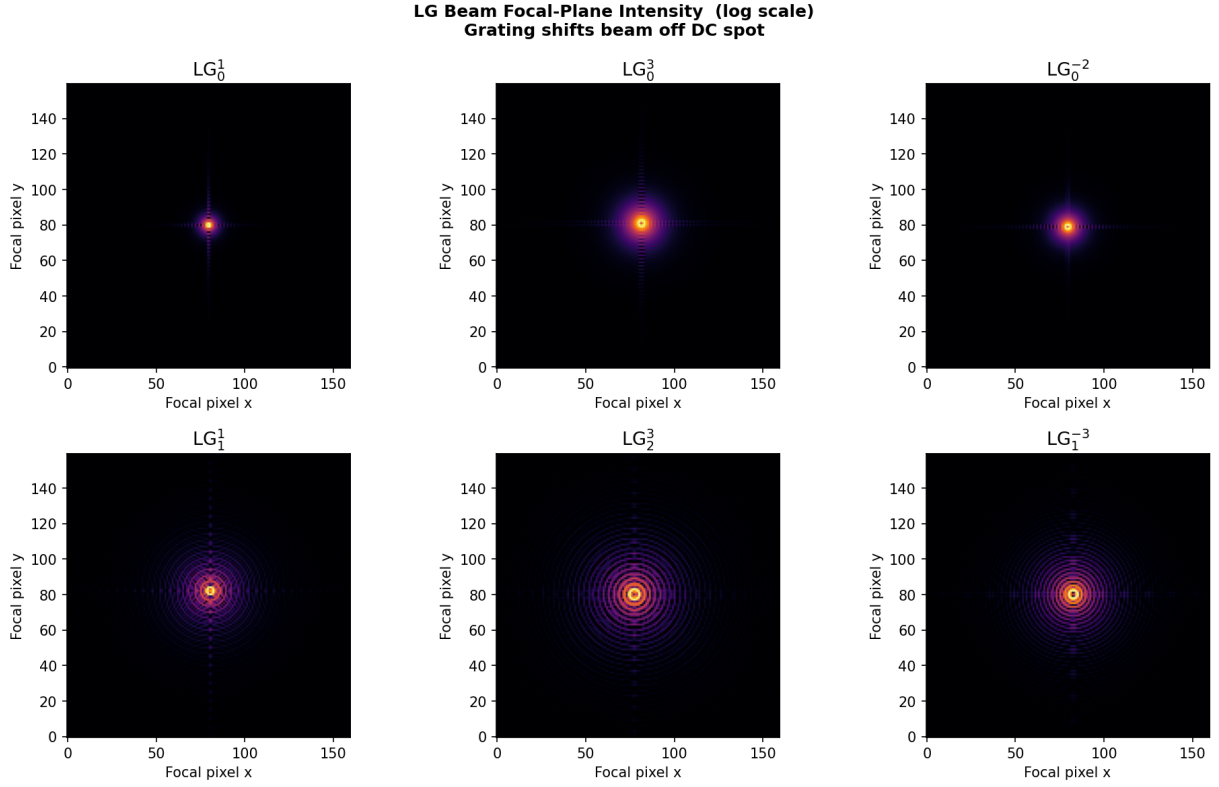


Figure 13: **Ex 11.3.1.** Focal-plane intensity for the same six modes. Doughnut radius grows with $|l|$; ring count with p .

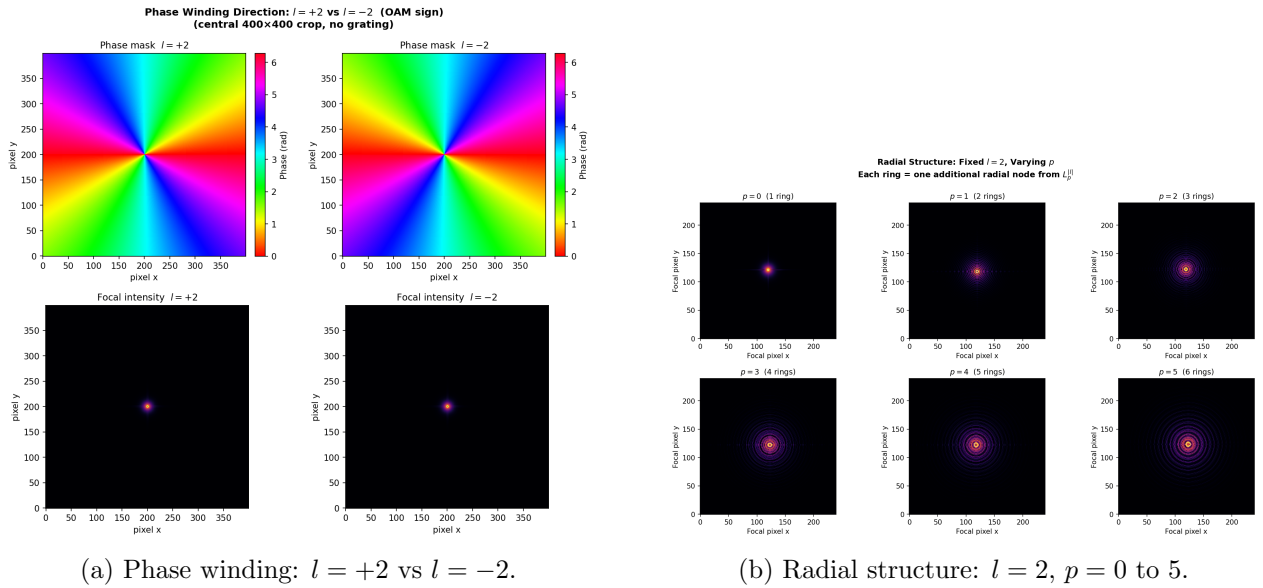


Figure 14: **Ex 11.3.1.** OAM sign visible in winding direction; radial index p adds one ring per unit.

4.2 Ex 11.3.2 — Hermite-Gaussian Beams

Binary SLM phase mask:

$$\phi(x, y) = \pi \Theta \left\{ H_{m_x} \left(\frac{\sqrt{2}x}{w_0} \right) H_{m_y} \left(\frac{\sqrt{2}y}{w_0} \right) \right\}. \quad (4)$$

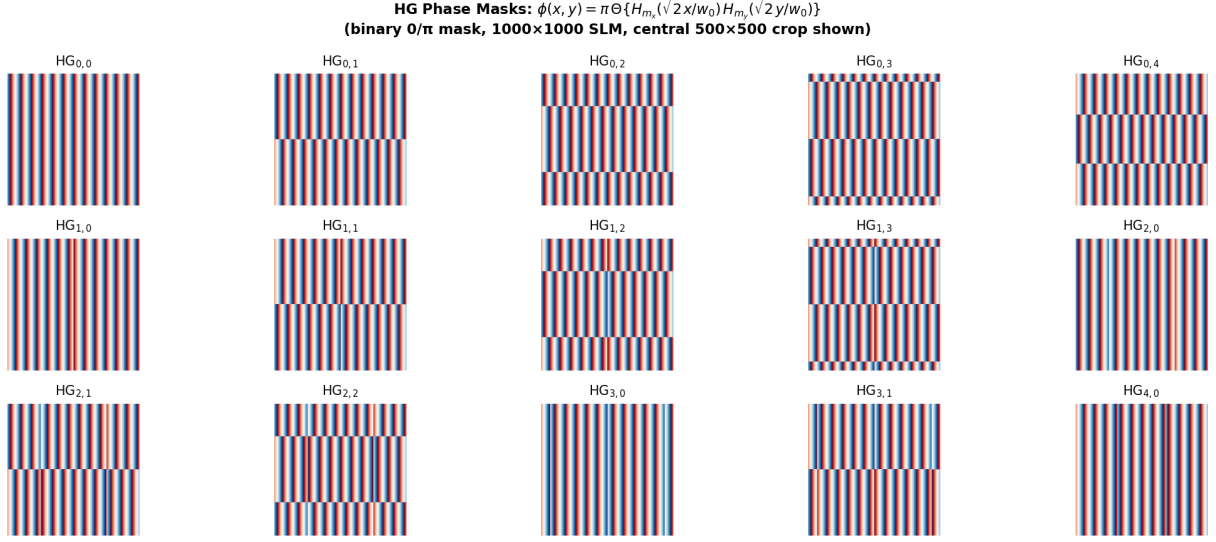


Figure 15: **Ex 11.3.2.** Binary phase masks for HG modes with $m_x + m_y \leq 4$.

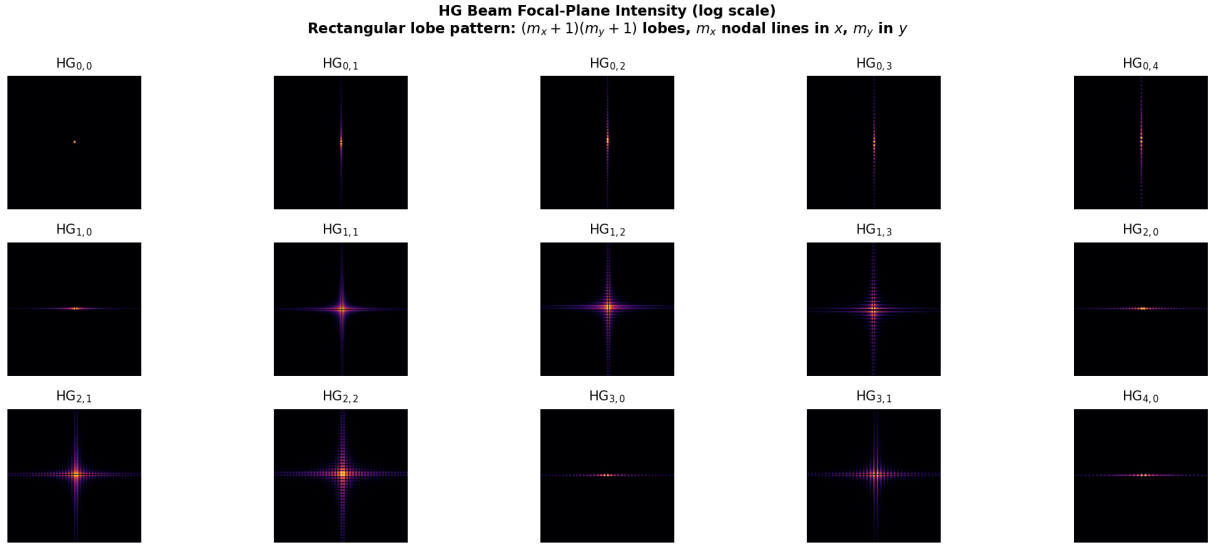


Figure 16: **Ex 11.3.2.** Focal intensity: $(m_x + 1)(m_y + 1)$ rectangular lobes.

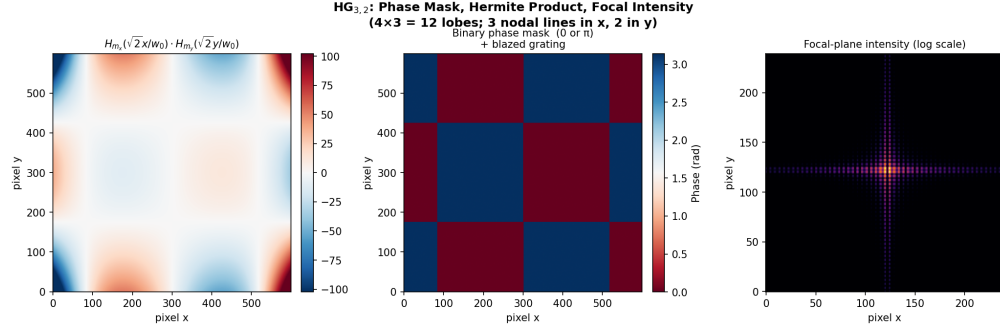


Figure 17: **Ex 11.3.2.** HG_{3,2}: Hermite product, binary mask, focal intensity ($4 \times 3 = 12$ lobes).

4.3 Ex 11.3.4 — Counter-Rotating Optical Traps

$\phi_{\text{SGL}} = \arg(\sum_n e^{i\phi_n})$ with charges $\pm l$. Opposite-OAM particles orbit in opposite directions.

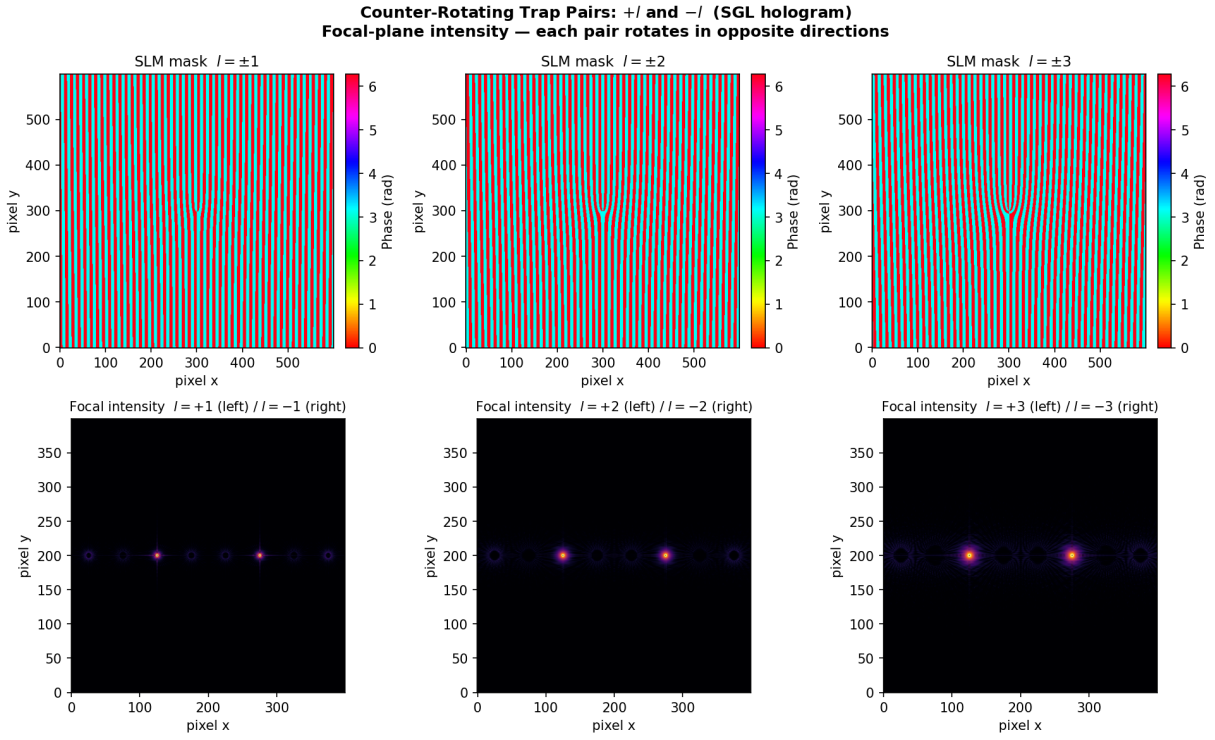
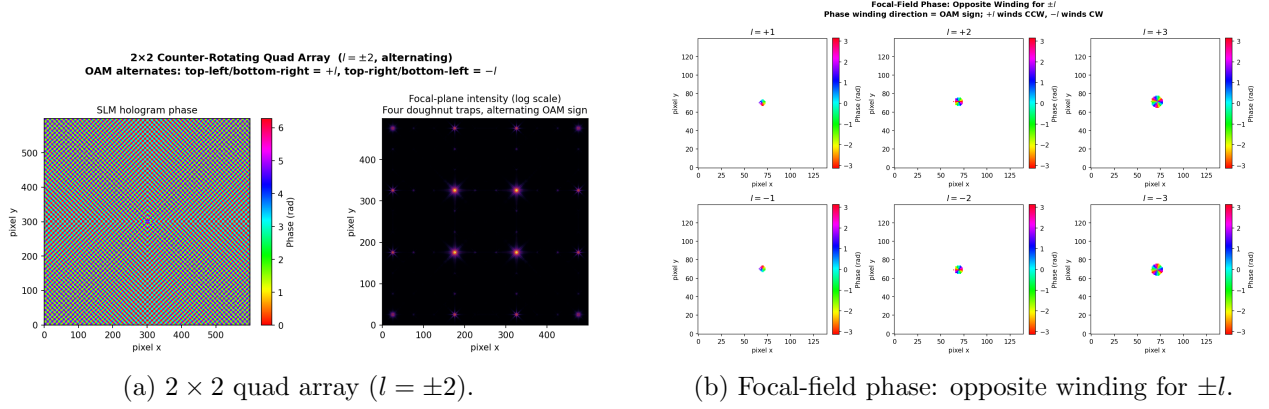


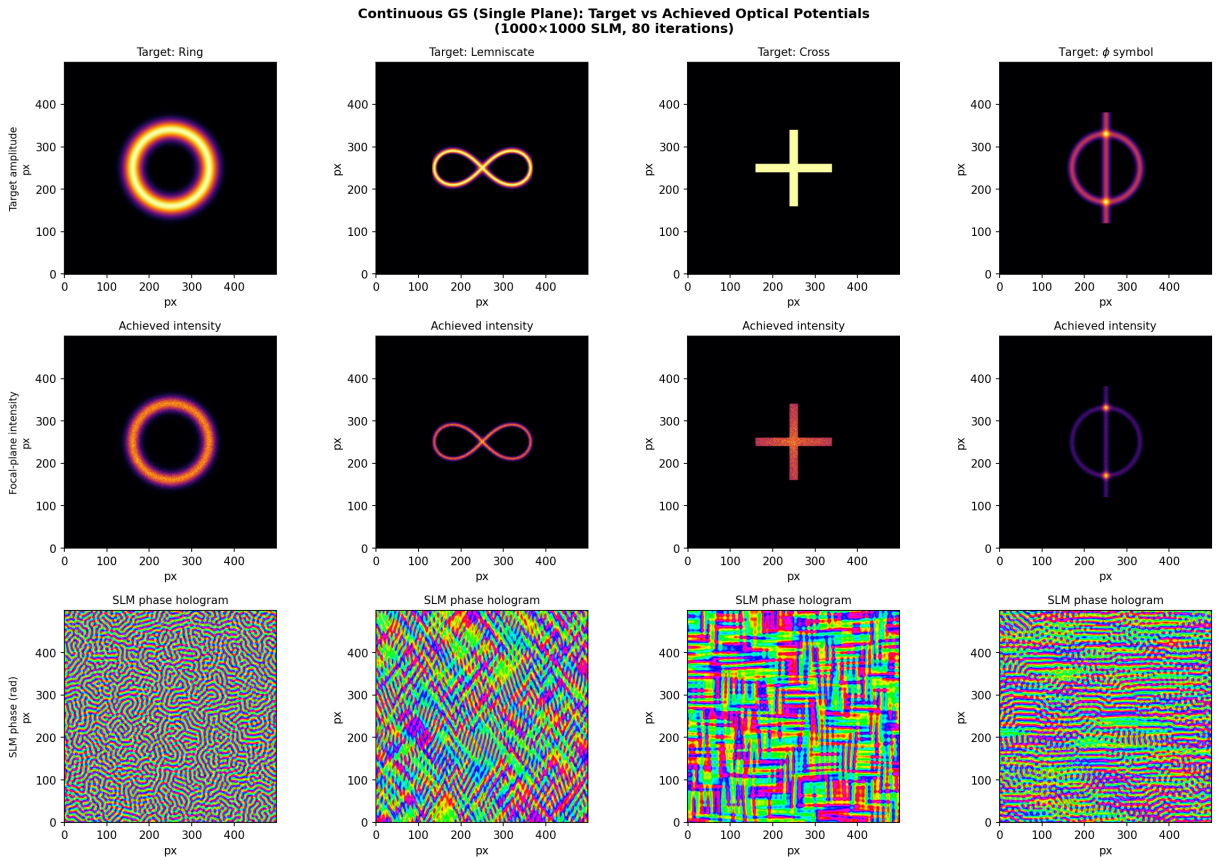
Figure 18: **Ex 11.3.4.** Counter-rotating pairs with $l = 1, 2, 3$.

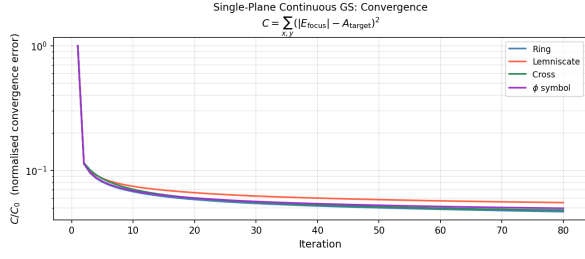
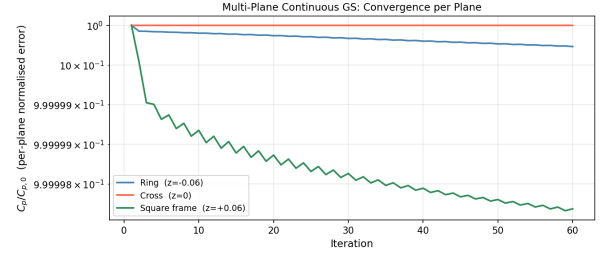
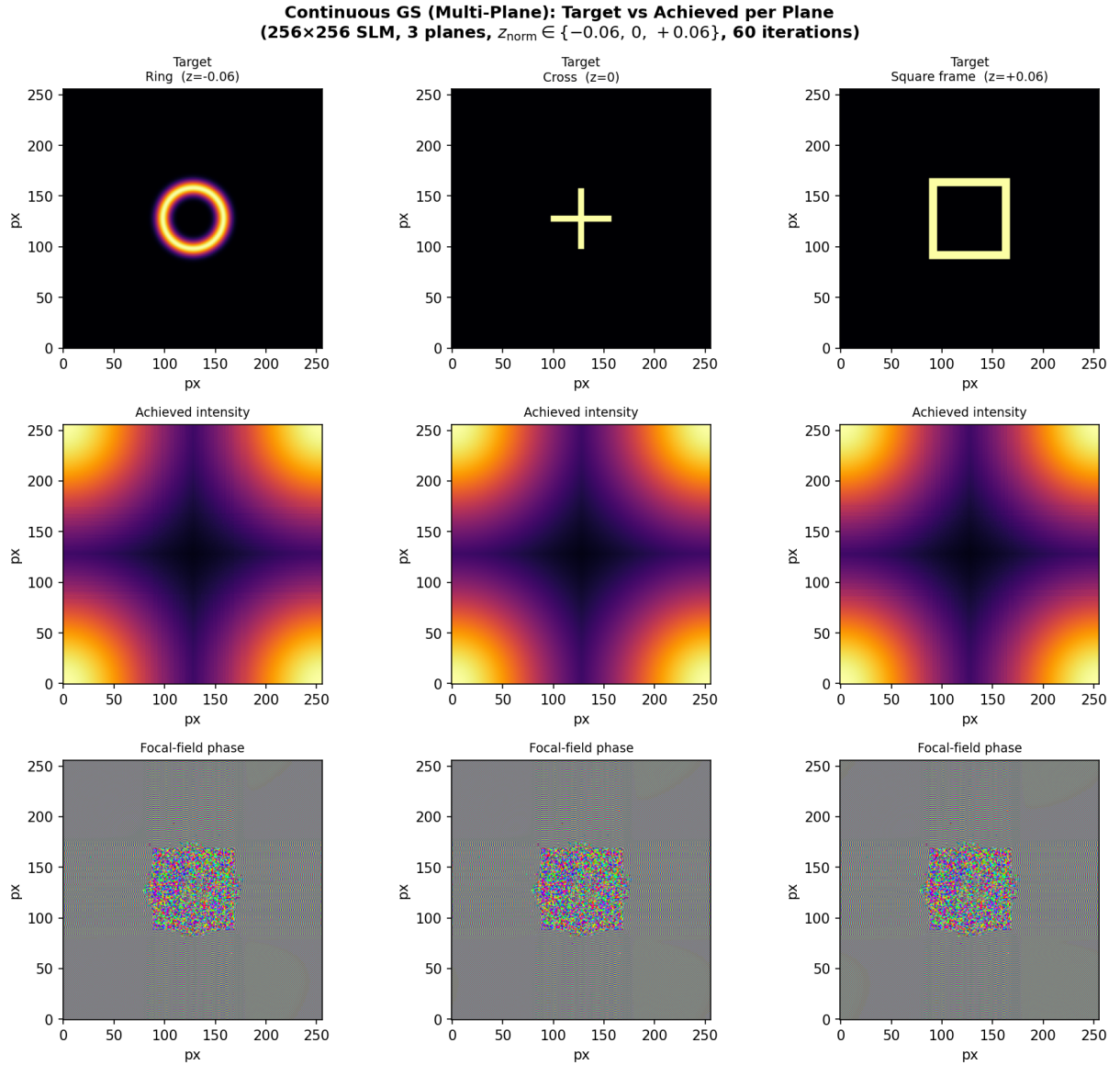
Figure 19: **Ex 11.3.4.** Quad array and phase winding maps.

Continuous Optical Potentials (11.4.1–11.4.3)

5.1 Ex 11.4.1 — Continuous Potentials via GS

Amplitude replacement applied over the full focal field. Target power-normalised via Parseval's theorem: $A_t = A_{\text{target}} \sqrt{\|E_{\text{SLM}}\|^2 M_x M_y / \|A_{\text{target}}\|^2}$.

Figure 20: **Ex 11.4.1.** Target, achieved intensity, and SLM phase for four continuous patterns on 1000×1000 SLM (80 iterations).

(a) Single-plane: C/C_0 vs iteration.(b) Multi-plane per-plane convergence (256×256).Figure 21: **Ex 11.4.1.** Convergence: most reduction in first 15 iterations.Figure 22: **Ex 11.4.1.** Multi-plane GS: target vs achieved at three planes.

5.2 Ex 11.4.2 — Proof: C is Non-Increasing Under GS

$C = \sum_{x,y} (|E_{\text{focus}}| - A_t)^2$. Each iteration consists of two alternating projections onto closed convex sets: step 2 projects onto $S_2 = \{E : |E| = A_t\}$ (setting $C = 0$); step 4 projects onto $S_1 = \{E : |E_{\text{SLM}}| = A_{\text{laser}}\}$. The alternating-projections theorem guarantees $C^{(k+1)} \leq C^{(k)}$. Parseval normalisation of A_t is essential; per-iteration rescaling breaks monotonicity.

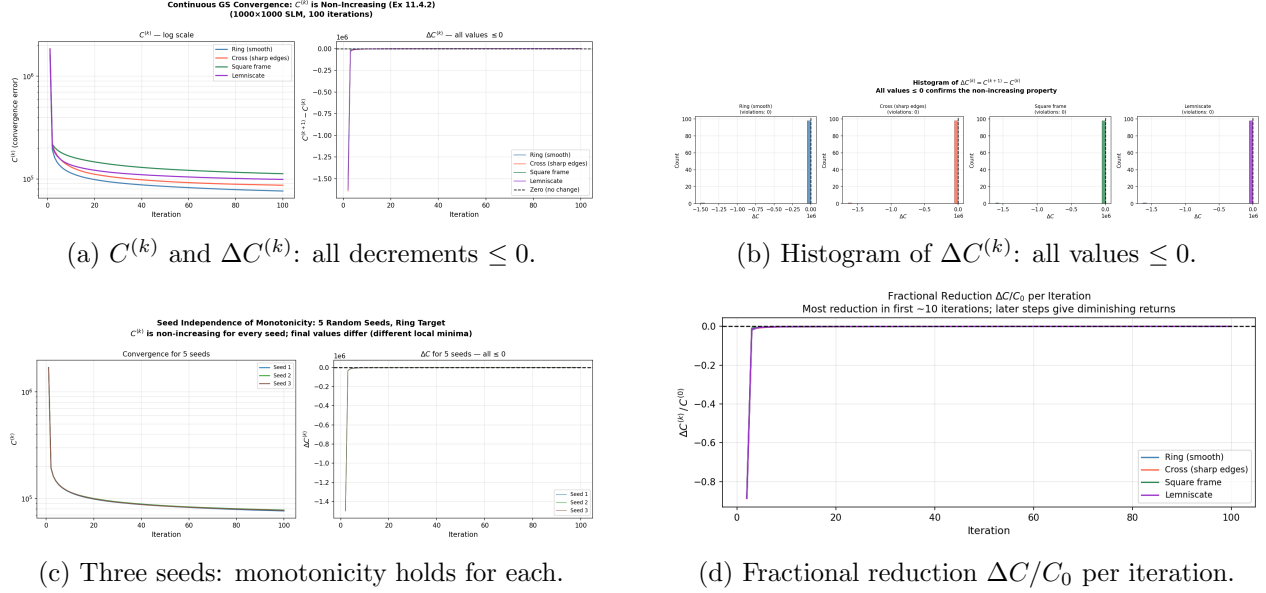


Figure 23: **Ex 11.4.2.** Zero violations across 4 targets and 3 seeds, 100 iterations, 1000×1000 SLM.

5.3 Ex 11.4.3 — Continuous Potentials via AA

$E_{\text{new}}(x, y) = (1 - a)E_{\text{current}} + aA_t e^{i\angle E_{\text{current}}}$. For continuous potentials, GS ($a = 1$) converges to lower C than AA ($a < 1$) at equal iteration count: the continuous projection leaves no discrete local traps for mixing to help escape.

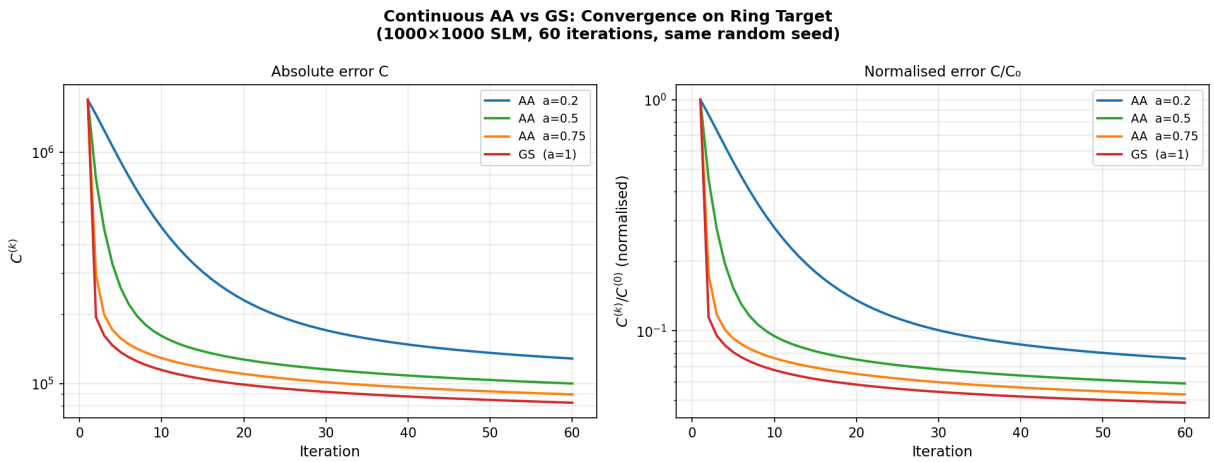
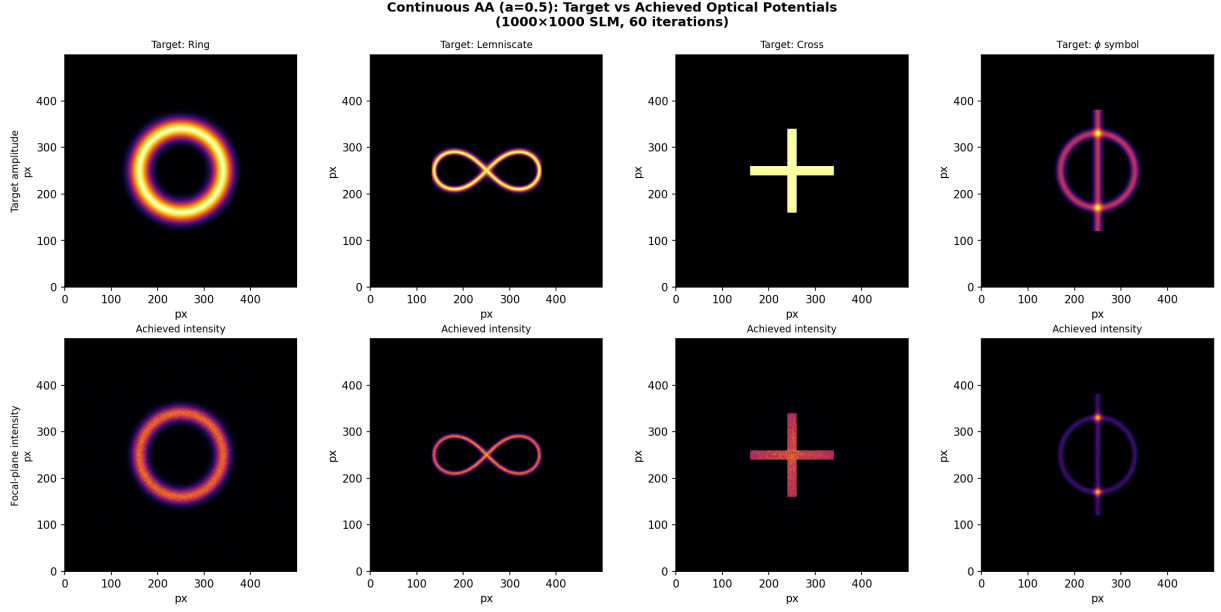
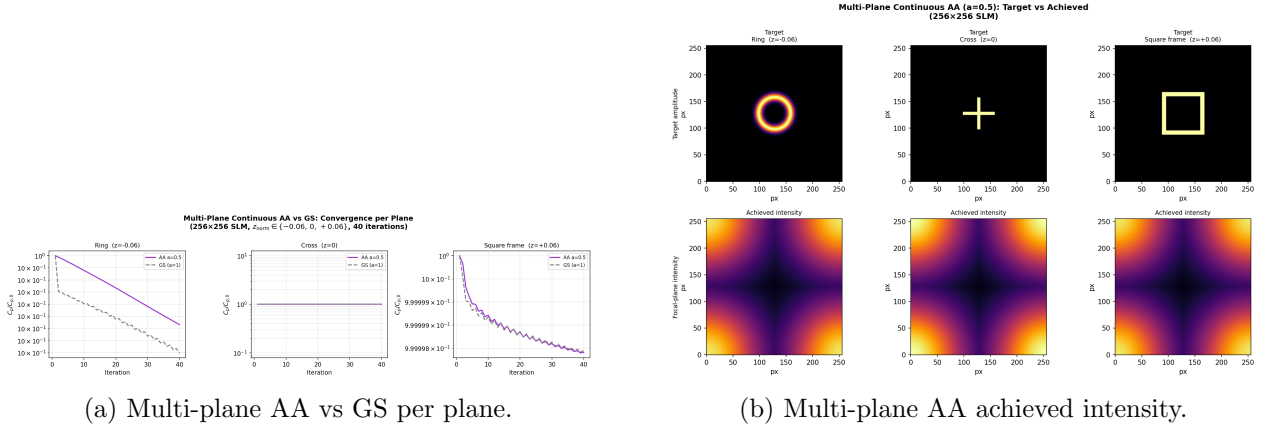


Figure 24: **Ex 11.4.3.** AA vs GS on ring target: GS wins for continuous potentials.

Figure 25: **Ex 11.4.3.** Target vs achieved for four patterns using AA $a = 0.5$.Figure 26: **Ex 11.4.3.** Multi-plane AA and GS both reduce C by $\sim 95\%$ from C_0 in 40 iterations.

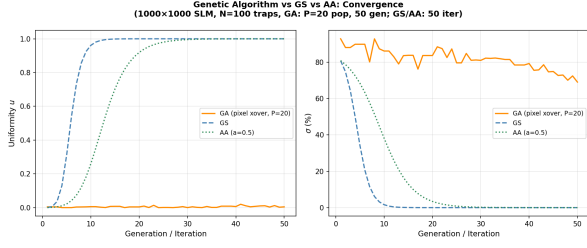
Problem Set (11.1–11.3)

6.1 Problem 11.1 — Genetic Algorithm

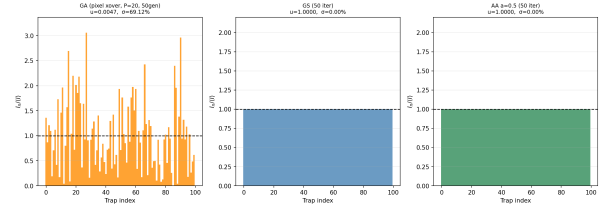
Fitness $F_{\text{gain}} = \langle I \rangle - w\sigma$ ($w = 0.5$); population $P = 20$; tournament selection; pixel-wise crossover; decaying mutation rate ($0.04 \rightarrow 0.004$); 50 generations.

Algorithm	u	σ (%)	$\langle I \rangle$	Time
GA ($P = 20$, 50 gen)	0.005	69	2.57	100 s
GS (50 iter)	1.000	0.00	0.010	6 s
AA $a = 0.5$ (50 iter)	1.000	0.00	0.010	6 s

GS achieves perfect uniformity $17\times$ faster. The GA under-penalises non-uniformity at $w = 0.5$ and its local-optima-escape advantage is not needed for the essentially convex GS landscape of this lattice.



(a) Convergence: GA vs GS vs AA.

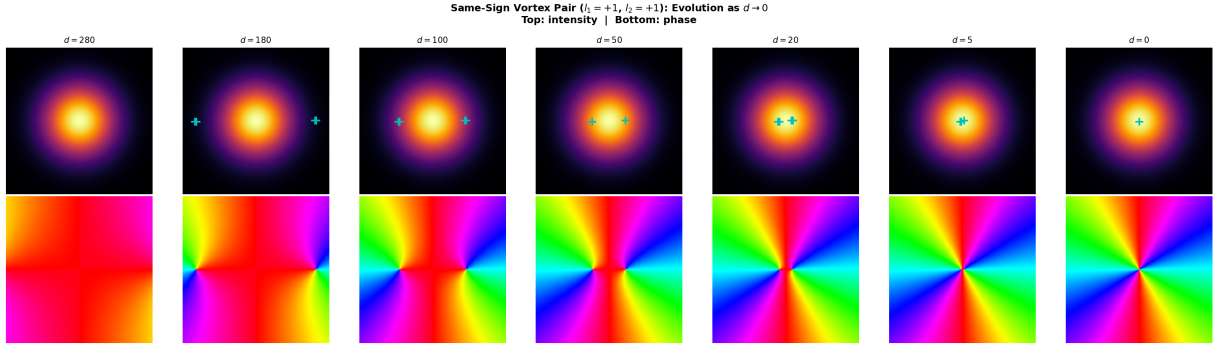
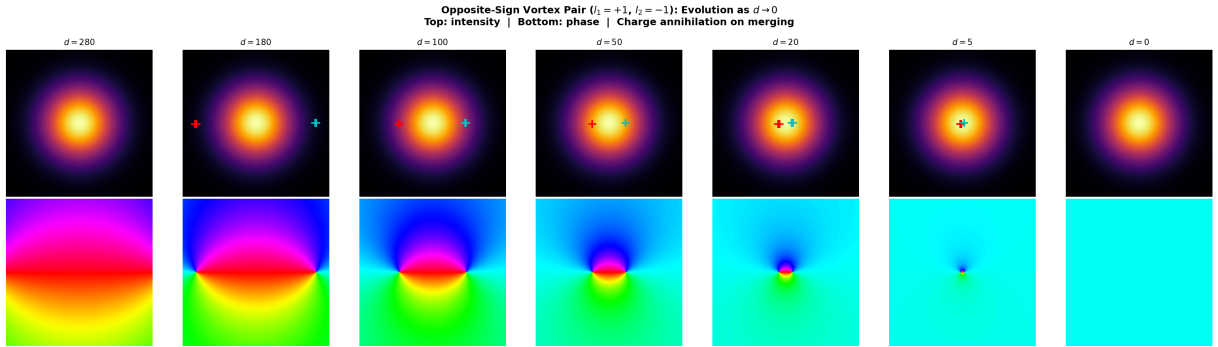


(b) Per-trap intensity: all three algorithms.

Figure 27: **Problem 11.1.** GA vs GS vs AA comparison.

6.2 Problem 11.2 — Optical Vortex Pair

$E(x, y) = E_G(x, y) \cdot e^{il_1 \arctan(y/(x-d))} \cdot e^{il_2 \arctan(y/(x+d))}$. Vortex positions detected via phase winding numbers on a grid.

Figure 28: **Problem 11.2A.** Same-sign pair (+, +): intensity and phase as d decreases from 280 to 0 px. Singularities merge at $d = 0$.Figure 29: **Problem 11.2B.** Opposite-sign pair (+, -): annihilation at $d = 0$.

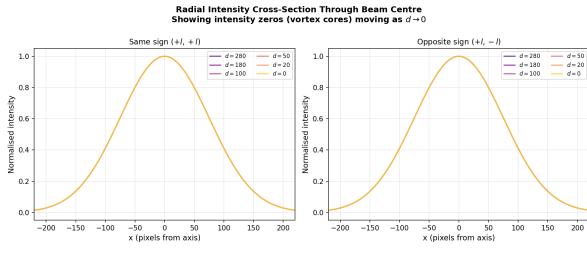
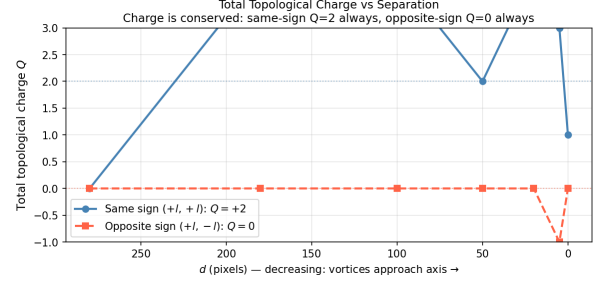
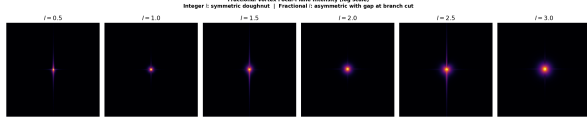
(a) Intensity zeros move as $d \rightarrow 0$.(b) Total charge Q : conserved throughout.

Figure 30: **Problem 11.2.** Topological charge is conserved: $Q = +2$ (same-sign) and $Q = 0$ (opposite-sign) throughout.

6.3 Problem 11.3 — Fractional Optical Vortices (Berry 2004)

$\phi_{\text{SLM}} = [l \arctan(y/x)] \bmod 2\pi$ for non-integer $l = n + \varepsilon$. The branch cut along $+x$ generates a chain of unit-charge vortices (Berry 2004), not a single fractional-charge singularity.

(a) Focal intensity: symmetric doughnut at integer l , asymmetric ring with gap at fractional l .

(b) Focal-field phase: unit-vortex chain along branch cut.

Figure 31: **Problem 11.3.** Fractional vortex structure (Berry 2004). Cyan $+$: charge- $+1$ vortex; red $\times$: charge- -1 vortex.

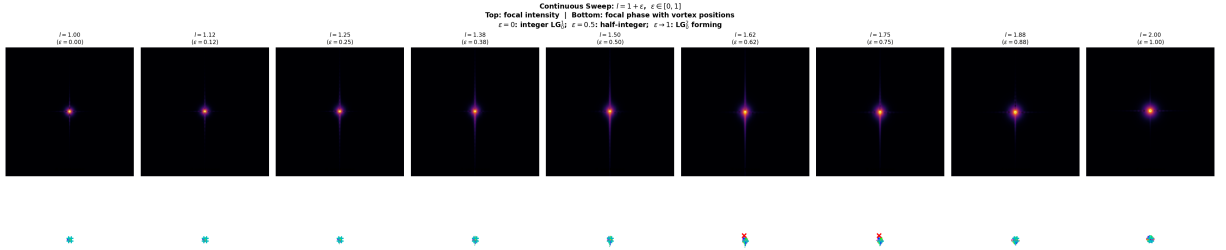


Figure 32: **Problem 11.3.** Sweep $l = 1 + \varepsilon$, $\varepsilon \in [0, 1]$. Vortices appear near centre at small ε and migrate outward.

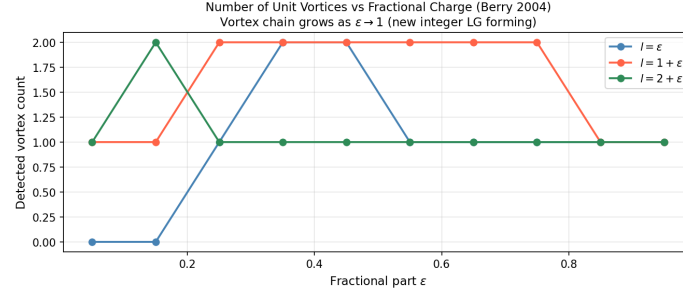


Figure 33: **Problem 11.3.** Detected vortex count vs ϵ . Consistent with Leach et al. (2004) and Lee et al. (2004).

Summary

Exercise	Key result	Grid	Script
11.2.1–11.2.10	Phase masks, RM, SGL, weighted	512^2	earlier chat
11.2.11 GS	$u = 1, \sigma = 0$ in 60 iter	1000^2	gs_algorithm.py
11.2.12 Sparse GS	FFT faster at $N = 100$	1000^2	gs_sparse.py
11.2.13 WGS	Uniformity via α -rescaling	1000^2	gs_weighted.py
11.2.14 3D GS	Multi-plane via Fresnel sum	256^2	gs_3d.py
11.2.15 AA	$a \geq 0.5$ matches GS at 60 iter	1000^2	aa_algorithm.py
11.2.16 DS	94% recovery (Cizmar 2010)	256^2	ds_optimal_focus.py
11.2.17 Zernike	Strehl $0.215 \rightarrow 1.000$	1000^2	zernike_correction.py
11.3.1 LG	Doughnut: $l\varphi + \pi\Theta\{L_p^{ l }\}$	1000^2	lg_beams.py
11.3.2 HG	Binary π -mask	1000^2	hg_beams.py
11.3.4 Counter-rot.	SGL + $\pm l$	1000^2	counter_rotating_traps.py
11.4.1 Cont. GS	Ring/lemniscate/cross/ ϕ	$1000^2/256^2$	continuous_gs.py
11.4.2 Proof	$\Delta C \leq 0$ (0 violations)	1000^2	gs_convergence_proof.py
11.4.3 Cont. AA	GS wins for continuous	$1000^2/256^2$	continuous_aa.py
P11.1 GA	GS $17\times$ faster	1000^2	genetic_algorithm.py
P11.2 Vortex pair	Charge conserved	600^2	optical_vortex_pair.py
P11.3 Frac. vortex	Unit-vortex chain	1000^2	fractional_vortex.py

References

- [1] P. H. Jones, O. M. Maragò, G. Volpe, *Optical Tweezers: Principles and Applications*, Cambridge University Press, 2015.
- [2] T. Cizmar et al., “In situ wavefront correction,” *Nature Photonics* **4**, 388, 2010.
- [3] M. V. Berry, “Optical vortices evolving from helicoidal phase steps,” *J. Opt. A* **6**, 259, 2004.
- [4] J. Leach et al., *New J. Phys.* **7**, 55, 2004.
- [5] W. M. Lee et al., *Opt. Lett.* **29**, 2467, 2004.
- [6] H. H. Bauschke, J. M. Borwein, *SIAM Review* **38**(3), 367, 1996.